

Social Media Content Automation Pipeline

n8n Cloud Workflow — Technical Walkthrough

Abdul Rehman Ali | Air University Kharian | abdulrehman.tp.786@gmail.com
linkedin.com/in/abdul-rehman-ali | abdulrehmanali.netlify.app

Project: A fully automated social media content pipeline built with n8n Cloud. The system fetches trending articles from multiple RSS feeds daily, filters for recency, checks for duplicates, generates viral LinkedIn posts using Gemini AI, delivers them to email, and tracks everything in a Google Sheets content calendar — with zero manual effort.

1 Workflow Architecture

The pipeline runs on a daily schedule trigger. Each node handles one specific responsibility in the content generation process.

Step	Node	Purpose
1	Daily Schedule	Fires every 24 hours automatically
2	Read RSS Sources	Reads feed URLs and categories from Google Sheets RSS Sources tab
3	Fetch RSS Feed	Fetches live articles from each RSS URL
4	Filter Recent Articles	Code node — keeps only articles published in last 24 hours, limits to 10 per run
5	Check Skip	Filter node — stops pipeline if no recent articles found
6	Fetch Existing Posts	Reads Content Calendar to get all previously generated post URLs
7	Duplicate Check	Code node — removes articles already in Content Calendar by URL matching
8	Check Duplicate Skip	Filter node — stops if all articles already processed
9	Generate LinkedIn Post	Basic LLM Chain v1.9 — sends article to Gemini AI for post generation
10	Prepare Post Data	Code node — pairs AI output with original article data using \$() reference
11	Save to Content Calendar	Appends generated post to Google Sheets with Ready status
12	Email Post to Self	Sends ready-to-publish post via Gmail with article context

2 Architecture Decisions & Rationale

Daily Schedule Trigger

Content marketing works best with consistent daily posting. A 24-hour schedule ensures fresh content is generated every day without manual intervention, building a sustainable posting habit automatically.

Multi-Source RSS Strategy

Reading from 4 different RSS sources (TechCrunch, MIT AI, HackerNews, VentureBeat) ensures content diversity across Technology, AI, Startup, and Business categories. This prevents the pipeline from over-indexing on a single source and keeps the content calendar varied.

24-Hour Recency Filter

Only articles published in the last 24 hours are processed. This ensures posts are always about current trending news rather than stale content, which is critical for LinkedIn engagement and relevance.

Duplicate Prevention via URL Matching

Before generating any post, the pipeline fetches all existing Content Calendar entries and cross-checks article URLs. This prevents the same article from being processed twice across multiple daily runs, keeping the content calendar clean and unique.

Gemini AI for Viral Post Generation

The prompt instructs Gemini to write posts with strong hooks, 2-3 key insights, business relevance, a thought-provoking closing question, and 3-5 hashtags. This structure follows proven LinkedIn content frameworks that maximize engagement and reach.

Email Delivery Instead of Auto-Posting

LinkedIn API requires a company page and 1-2 weeks approval. Email delivery allows human review before publishing — this is actually a feature, not a limitation, as it prevents accidental publication of AI-generated content that may need refinement.

\$() Node Reference for Data Pairing

Same solution as the Lead Generation project — LLM Chain drops input fields from output. Prepare Post Data node uses `$("Check Duplicate Skip").all()` to re-fetch original article data and pair it with AI output by array index.

3 Assumptions Made

RSS Feed Availability

- All RSS feed URLs remain active and return valid XML — no feed validation is performed.
- RSS feeds publish at least one new article every 24 hours — if not, the pipeline skips gracefully.
- Article titles and summaries in RSS feeds are sufficient context for Gemini to generate quality posts.

Content Quality

- Gemini generates professional, brand-appropriate content — no content moderation layer is applied.
- Generated posts are reviewed by the human before publishing — email delivery enables this review step.
- LinkedIn hashtags suggested by Gemini are relevant to the target audience.

Infrastructure

- Google Sheets OAuth2 credentials remain valid for daily automated reads and writes.
- Gmail OAuth2 credentials remain valid — no token refresh issues during scheduled runs.
- Gemini API daily limit accommodates the number of articles processed per day.

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Error Handling & Edge Cases

Scenario	How It Is Handled
No recent articles found	Filter Recent Articles returns skip flag — Check Skip node stops pipeline gracefully
All articles already processed	Duplicate Check returns skip flag — Check Duplicate Skip stops pipeline
RSS feed URL is down	n8n continues with remaining feeds — one failed fetch does not stop the workflow
Gemini API rate limit	Failed items are skipped — re-run workflow or wait for next daily trigger
LLM Chain drops input fields	Prepare Post Data uses \$() node reference to re-pair data by array index
Gmail delivery failure	Post is still saved to Content Calendar — can be retrieved manually from sheet
Duplicate articles across sources	URL-based duplicate check catches same article appearing in multiple RSS feeds
Article summary too long	Summary is truncated to 500 characters before sending to Gemini to avoid token limits

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Known Limitations & Future Improvements

Current Limitations

- **No LinkedIn auto-posting:** Requires LinkedIn API approval — email delivery used instead.
- **Gemini free tier:** 20 API calls/day limits the number of posts generated per run.
- **No engagement tracking:** Cannot measure likes, comments, or reach of published posts.
- **No topic filtering:** All RSS articles are processed regardless of relevance to niche.
- **Single language:** Only generates English posts — no multilingual support.

What I Would Improve With More Time

- **LinkedIn API integration:** Auto-publish directly to LinkedIn after approval.
- **AI relevance scoring:** Filter articles by relevance score before generating posts.
- **Engagement analytics:** Track post performance and optimize content style over time.
- **Multi-platform support:** Generate Twitter/X threads and Instagram captions alongside LinkedIn posts.
- **A/B testing:** Generate 2 post variants and track which style performs better.

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Technology Stack

Tool	Usage	Reason
n8n Cloud	Workflow orchestration	Visual builder with code nodes — handles complex multi-step automation
Google Gemini AI	LinkedIn post generation	Strong language model, produces human-sounding viral content
RSS Feed Reader	Article fetching	Native n8n node — reads multiple RSS feeds in one pass
Google Sheets	RSS source management + content calendar	Easy to add new RSS sources, full post history visible
Gmail OAuth2	Daily post delivery	Reliable delivery, posts arrive ready to copy-paste to LinkedIn
JavaScript (Code nodes)	Filtering, deduplication, data pairing	Full control over recency filter, URL matching, and LLM data fix
Luxon (built-in n8n)	Timestamp formatting	Pakistan timezone timestamps for content calendar entries

Abdul Rehman Ali — AI Automation Builder & BSCS Student, Air University Kharian
abdulrehman.tp.786@gmail.com | linkedin.com/in/abdul-rehman-ali | abdulrehmanali.netlify.app